UNIVERSITY OF TEHRAN
Electrical and Computer Engineering Department
Object-Oriented Modeling of Electronic Circuits, Spring 1405
Computer Assignment 6
SAYAC ISS and custom instructions – Covering materials through Class 20

Name **Date**

Description:

In mail-drop drone navigation, GPS plays an important role. Using a GPS module, the drone's location can be obtained as a pair of X-Y coordinates. However, what is needed for navigation is distance and angle, which can be computed by the flight management system. In this assignment, you are going to implement this application and accelerate it using a Custom-Instruction on the SAYAC embedded system.

Part 1: Software Implementation: In this part, you are going to implement a software code to calculate the distance, and the angle based on the current location and the destination location.

The procedure is as follows:

Step-0: You have the current location (x_1, y_1) and the destination location (x_2, y_2) as four integer values in memory.

Step-1: Read the data and compute Euclidean distance using **"Equation 1".**

Equation 1. $(x_2 - x_1,\ y_2 - y_1) = (\Delta x, \Delta y)$

Step-2: Compute the distance using **"Equation 2"**, and the angle using **"Equation 3"**.

$$\boldsymbol{Equation\ 2.}\quad distance = \sqrt{((\Delta x)^2 + (\Delta y)^2\ )}$$

$$\boldsymbol{Equation\ 3.}\quad angle = \tan^{-1}\frac{\Delta y}{\Delta x}$$

Step-3: Write the result in the memory.

To do:

1. Write the C program for calculation of *distance* and *angle*.
2. Compile the C program you developed using out SAYAC Home Made Compiler.
3. Use SAYAC ISS for running the compiled code.
4. In Cycle-accurate SAYAC ISS, find the execution time for calculation of the distance and angle.

Part 2: Custom Instructions: To accelerate execution and reduce the instruction count, custom instructions can be added to the processor core of an embedded system. Here, you are to add custom instructions to the SAYAC ISA for distance and angel calculations.

To do:

1. Design two hardware units for distance and angle computation and show their block diagrams. Details of hardware are not necessary.
2. Based on your hardware, find the approximate time in terms of clock cycles.
3. Add two instructions to SAYAC ISA for distance and angel computing with above timing.
4. Modify the software code you developed in Part 1 to use custom instructions.

5. Report the execution time of above code.

Part 3: Conclusions: Here, you are to compare the results for an embedded system with and without custom instructions.

To do:

1. Compare the execution time.
2. Describe the pros and cons of using custom instructions. (Don't forget: there is no free lunch.)

Deliverables:

A complete report containing answers to all parts of each question. Your report should include enough design illustration, description, actual data, and output justification. Note that your reports should be well-organized.

Attention:

Make a *PDF* file of your report and submit it to the course site. Also, compress all files and documents mentioned in the *Deliverables* section into a *zip* file and upload the generated file. The name of the zip file must be in this format "***YourFirstName-YourLastName-HW6***". In addition, use exactly this phrase "***Submitting HW#6***" as the subject of your email.